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Embedded finance and FinTech disappearance

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Abstract

With reference to Elon Musk's FinTech strategy for X, the social media platform formerly known as Twitter, this essay critically interrogates the evolution of North American and European FinTech economies toward what is typically called 'embedded finance'; that is, the technological integration of monetary and financial services into discrete social interactions and economic transactions by nonfinancial companies. We argue that embedded finance furthers the disappearance of FinTech as an evident market domain of technologically facilitated monetary and financial relations. Specialist FinTech startup intermediaries are receding into the background of an institutional and digital landscape shaped by strong monopolization tendencies. FinTech economies are increasingly dominated by major platform firms with the assistance of banks. Relatedly, FinTech services have become ubiquitous to the extent that they are taken for granted by people who are configured as platform users that are ripe for rent capture, rather than as sovereign consumers searching for products. The disappearance of FinTech should not be confused with its demise, however. Disappearance is the fullest expression of the transformative appearance of FinTech in people's everyday monetary and financial lives over the last quarter century.

Keywords: BigTech; embedded finance; FinTech; monopolization; user subjects

Introduction: 'The everything app'

In November 2022, shortly after the US\$44 billion private acquisition of Twitter by Elon Musk's X Holdings, he announced that FinTech would be key to the future of the loss-making social media platform (*The Economist*, 2022). Twitter's already shaky economic prospects appeared to have worsened during the two weeks after Musk took ownership, in part because he was busy making more than half of the workforce redundant. At a meeting on Twitter's live audio facility to placate the advertising clients who are the platform's principal source of revenue, Musk took the opportunity to set out his vision for the firm. Announcing for the first time his intention to rebrand Twitter as 'X', Musk stated that 'people will not only be able to post their thoughts but also pay for goods, make phone calls, or watch movies' (in Conger and Mac, 2024, location 233). Twitter would become what he called 'the everything app' (Hsu and Conger, 2022; Murphy and Alim, 2022).

The announcement was short on specifics, although in the preceding week Twitter had paved the way to becoming a payments processor in the US, filing necessary registration paperwork with the Treasury Department's Financial Crimes Enforcement Network (FinCEN). Twitter was rebranded in July 2023 and, according to the company's website,

between late 2022 and early 2025, X Payments LLC secured payment licences to operate in 41 US states. The launch of the newly named 'X Money' in the US was planned for 2025, although it is unclear at present whether the roll-out will meet this deadline because a host of factors have contributed to delays, including: Musk's short-lived position as head of the Department of Government Efficiency (DOGE) in Donald Trump's second Presidential administration; the administration's decisions to impose tariffs on US trading partners and disempower financial regulatory agencies which have destabilized financial markets and introduced new uncertainties into FinTech economies (*The Economist*, 2025; Gilisson, 2025); the July 2025 resignation of Linda Yaccarino, the CEO who was actually responsible for transforming X into 'the everything app' (Thadani and Yang, 2025); and, perhaps most significantly, repeated refusals by the New York State Department of Financial Services (NYSDFS) to concur with regulators from other states and grant X Money regulatory approval, not least because of concerns about compliance with anti-money laundering and fraud requirements (Wayt, 2025).

Shoring up Twitter's investor and banking relations was a further reason for Musk's 'everything app' announcement. Whether FinTech services are ultimately offered on the X platform was rather less important at the time of the announcement than the company offering up the possibility that they would be, at some indeterminate point in the not-too-distant future, with all the additional income and returns on investment that this would presumably deliver. Despite a personal net worth of over US\$200 billion in late 2022, Musk enlisted a group of around 90 private equity investors to raise the funding for his acquisition of Twitter. They included doyens of the Silicon Valley tech scene, US venture capitalists, family offices from across the globe, a member of a Gulf state royal family, an American rap music performer, and close to 30 separate units of Fidelity, the asset management firm (Bickerton, 2024). The purchase also rested on complex capital raising structures that were hastily stitched together at the last minute to fund the deal. Musk's original intention had been to sell a portion of his Tesla stock to buy Twitter. However, as negotiations around the Twitter deal dragged on during 2022, Tesla's stock price continued to fall on news of Musk's plans and undermined the viability of this strategy. Instead, the acquisition was made with the help of US\$12.5 billion worth of loans advanced by seven major US banks. But in the macroeconomic conditions of 2022, characterized by financial uncertainty, rising inflation and higher interest rates, these banks were unable to entice investors to buy securitized repackages of the loans prior to the acquisition.¹

Musk sought to reframe Twitter as a viable proposition against the backdrop of a near ten-year cycle of global venture capital and private equity investment in which FinTech startups had proven to be a principal asset class. At the peak of the cycle in 2021, total annual global venture investment in FinTech startups reached US\$140 billion, 500 percent higher than in 2014. By the end of 2021, as the global population of 'unicorns' across all economic sectors reached 959, more than one-quarter (250) of privately-owned firms valued at over US\$1 billion were FinTech startups (Langley and Leyshon, forthcoming). Major banks, technology firms, and nonfinancial companies were also ramping up investment in their FinTech operations during this period. For instance, in 2022 alone, JPMorgan Chase, the largest US bank by assets, deposits, and market capitalization, spent one-third of its staggering US\$12 billion budget for new technology on the development of its main smartphone banking app, Chase Mobile (JPMorgan Chase, 2023). In the terms of a media report on Musk's 'everything app' plan for Twitter, then, FinTech was 'certainly in the Zeitgeist' in Silicon Valley and on Wall Street (Harrison, 2022). Revealingly, in March 2025, Musk performed a similar pivot to once again financially reengineer X (Murphy, 2025), this time repositioning the platform in the context of the spiraling global investment fad for AI. As a result, X is now owned by its former subsidiary, xAI.

In this essay, we take Musk's 'everything app' strategy for X as a provocation to critically interrogate the evolution of FinTech economies toward what is typically called

‘embedded finance’.² Since around the turn of the decade, the notion of embedded finance has become increasingly pronounced in North American and European FinTech economies, adopted by venture capitalists (e.g., Harris 2019a, 2019b), management consultants (e.g., Townsend, 2021), and media commentators (e.g., Sieber and Guibaud, 2022). In practice, embedded finance can be relatively extensive. In the US, for example, Walmart is currently developing a full range of monetary and financial services as part of its mobile app strategy, Walmart One, launched in 2021 (Shelvin, 2021). Embedded finance can also be limited to, for instance, a peer-to-peer (P2P) payment service incorporated into a social media platform, such as X, a buy-now-pay-later option on a merchant’s website, or an insurance policy option integrated into an online car rental system. As these examples indicate, the notion of embedded finance refers to the technological integration of monetary and financial services into discrete social interactions and economic transactions by nonfinancial companies, most obviously, at a point of sale (PoS). As leading advocates Scarlett Sieber and Sophie Guibaud (2022: 3) excitedly proclaim, embedded finance ‘equips technology companies, brands, and retailers with the ability to provide a banking and payments experience to their customers in a seamless, convenient, and authentic way . . . when they need it most’, while ‘From the consumer perspective, . . . the financial transaction becomes naturally integrated into what you are doing to the point it feels invisible’.

With reference to Musk’s FinTech vision for X, this essay argues that the rise of embedded finance furthers the disappearance of FinTech as an evident market domain of technologically facilitated monetary and financial relations. Under embedded finance arrangements, disruptive specialist startup intermediaries that were previously foregrounded in North American and European FinTech economies are receding into the background. The institutional and digital landscape of FinTech economies is shaped by strong monopolization tendencies and increasingly dominated by major platform firms with the assistance of banks. Relatedly, FinTech services are becoming ever-present and ubiquitous to the extent that they are taken for granted by people who are configured as platform users rather than as sovereign consumers.

FinTech disappearance does not spell the end of FinTech economies, however. Quite the contrary. As Jean Baudrillard (2009: 9) contends, the ‘question of disappearance’ is, more broadly, ‘not exhaustion, extinction or extermination’.³ There is no ‘linear trajectory . . . from appearance to disappearance’ (Baudrillard, 2009: 20). Embedded finance furthers the disappearance of all but the most established of FinTech startup firms and consumer markets for monetary and financial services. Yet, disappearance is also a marker of the appearance of today’s FinTech economies. FinTech intermediation and rent capture are increasingly dominated by the platform enclosures of the leading nonfinancial firms. Everyday monetary and financial relations are transformed into regularized relations of use. Indeed, as we will argue, the description of people as users is a powerful register of the intense reordering of the social relations and subjectivities of money and finance that is fashioned by FinTech and furthered by embedded finance.

From disruption to monopolization

In announcing his ‘everything app’ plan for Twitter in November 2022, Elon Musk invoked his early experience as a FinTech entrepreneur. Back in 1999, Musk co-founded X.com, declaring that ‘What we’re going to do is transform the traditional banking industry’ (in Lepore, 2025). His initial FinTech venture morphed into PayPal in 2001 and he profited handsomely when the firm was acquired by eBay in 2002. Musk bought back the single-letter domain name in 2017 (Bradshaw, 2022). Until relatively recently, especially in North America and Europe, FinTech economies were strongly associated with Silicon Valley

startups, such as PayPal, or what *The Economist* (2015) magazine memorably described as ‘the magical combination of geeks in T-shirts and venture capital’. This early narration of FinTech economies followed from the ‘California ideology’ of Silicon Valley innovation that first came to prominence during the ‘new economy’ around the turn of the millennium (Barbrook and Cameron, 1996). Accordingly, during the 2010s, FinTech economies were often described in terms of Clayton M. Christiansen’s (1997) influential notion of ‘disruption’. Innovative FinTech startups appeared to be ‘unbundling’ and ‘disintermediating’ the business of banking and deepening markets for financial services.

By the end of that decade, however, there was a dialing down of the disruption discourse in North American and European FinTech economies. The notion of embedded finance began to gain traction because, as one of its leading advocates put it, ‘moving forward ... there won’t be “fintech” companies as such’ (Harris, 2019a). There was also a realization around this time that the major US BigTech conglomerates were becoming significant players in FinTech economies. Amidst the ‘tech lash’ and associated drive to tailor new anti-trust legal regimes for BigTech firms (Cioffi, Kenney, and Zysman, 2022), financial regulators turned their attention to the FinTech operations of the major technology companies (Bains, Sugimoto, and Wilson, 2022; Borio et al., 2022; Carstens et al., 2021). In truth, BigTech had been present in FinTech economies across the globe for considerably longer. In China, for instance, Alibaba and Tencent were pivotal to FinTech economies from the early 2000s and evolved by giving monetary and financial services a pivotal place in their ‘super apps’ (Jia, Nieborg, and Poell, 2022). Major nonfinancial companies had for some time also played leading roles in FinTech economies elsewhere in the world. Telcos were central from the outset to FinTech economies in sub-Saharan Africa, for example, whilst ride-hailing companies were initially key FinTech players in the major cities of Southeast Asia.

Notably, then, while pointing to his earlier record as a FinTech founder, Musk also cited Tencent’s WeChat as his inspiration for integrating FinTech services into Twitter. WeChat is often highlighted as a leading use case in Western commentaries and consultancy knowledge on embedded finance.⁴ WeChat Pay is one of 200 or so mini software applications nested within the WeChat super app which provides around 1.3 billion users with access to news, medical, transport, food delivery, and monetary and financial services (Goggin, 2021). In online and mobile payments, in particular, FinTech in China is a duopoly dominated by the WeChat and Alipay super apps, the latter part of Ant Group and owned by Alibaba. Moreover, at the time Musk unveiled his WeChat inspired plan for Twitter, the US BigTechs were also stepping up their strategies to more fully assume positions at the head of FinTech economies. Apple was pursuing an especially aggressive FinTech growth strategy in the US, for example (Foroohar, 2023; McGee and Franklin, 2023), while Meta was busy adding WhatsApp Pay to its social messaging platform in India and Brazil (Murphy and Findlay, 2020; Murphy, 2020).

Compared to most of the major US technology companies, Musk’s Twitter arrived very late to the FinTech party in November 2022 (Perez, 2022). Nonetheless, as has been the case for its US competitors, the firm’s FinTech makeover will rest upon leveraging its social media platform and data assets. The increasingly dominant position of leading technology firms in FinTech economies is partly the result of general tendencies toward the concentration and centralization of monopoly capital, as FinTech startups are squeezed out by larger competitors who also acquire them in full or act as their minority investors. Core to the specific monopolization processes shaping FinTech economies and the rise of embedded finance, however, is exclusive ownership of the platformed means of connection and circulation in digital capitalism more broadly (Manokha, 2018; Narayan, 2023; Vasudevan, 2022), alongside corporate control of the data-derived means of knowledge production (Birch, 2023; Rikap 2021, 2023). While digital platform technologies and data analytics have created opportunities for disruptive FinTech startups to compete

with incumbent banks and financial institutions, proprietorial monopolies have provided the behemoths of digital capitalism with powerful competitive advantages as they have moved into FinTech. And, growing awareness amongst regulators of the challenges posed by BigTech FinTechs has not prompted new rules and restrictions, apart perhaps from the quashing of Meta's plans to issue a stablecoin called Libra (later Diem) that would have been integrated into the digital payment technologies available to two billion people on the Facebook platform (Murphy and Stacey, 2022).

For X Money to go live, the firm will also require partnerships with institutions of different kinds. Such partnerships will be in addition to strategic agreements with Amazon Web Services and Google Cloud that, since 2018, have increasingly provided the underlying infrastructure and data storage capacities of the X platform. In January 2025, for example, VISA became X's first official FinTech partner (*The Paypers*, 2025). Connecting X users to their bank accounts via its card payment network, VISA will provide the digital wallet technology for X Money. This is largely typical of embedded finance arrangements wherein nonfinancial firms partner with financial institutions. In this instance, the role played by VISA is similar to roles which are often performed by business-to-business (B2B) FinTech startups. B2B firms have become increasingly prominent in FinTech economies over time, relative to their ostensibly disruptive business-to-consumer (B2C) counterparts.

For banks, moreover, embedded finance partnerships often provide a further opportunity to respond to the competitive challenge of FinTech. Leading banks have frequently been able to defend their incumbent positions by technologically remaking their own intermediary operations. FinTech economies have thus been shaped by the monopolization tendencies of both financial capitalism and digital capitalism. Yet even smaller banks have also been able to expand into FinTech economies by acting as so-called 'Banking-as-a-Service (BaaS) partners in the various embedded finance operations of nonbanks (Mikula, 2024). As such, while the concentration and centralization of banking capital has shaped the institutional landscape of FinTech, so too has the exclusive possession of prized deposit-taking banking licenses. By leveraging their licenses, banks of all kinds benefit from a unique competitive advantage in embedded finance and FinTech economies more generally.

With the rise of embedded finance, then, once feted disruptive B2C FinTech startups are certainly disappearing. As is the case in many sectors, the major technology companies have achieved positions of dominance, thanks largely to the distinctive proprietorial monopolization processes of digital capitalism. The monopoly tendencies of financial capitalism that feature exclusive legal and regulatory provisions have also enabled banks to develop FinTech operations of their own, including BaaS strategies and the performance of essential supportive roles in embedded finance partnerships. B2C FinTech firms now tend to occupy less prominent and visible positions in the changing institutional and digital architecture of FinTech economies. FinTechs are no longer to the fore in FinTech.

From consumers to users

According to Musk's FinTech vision for Twitter, the platform's 100 million daily users in the US will first verify their identities. Next, to pay for content and make P2P transfers, they will connect their bank accounts to the digital wallet provided by VISA which is due to be integrated within the platform. Later, Twitter will offer what Musk loosely described as 'an extremely compelling money market account' and 'debit cards, checks and whatnot' (in Webber, 2022). Indeed, shortly before Twitter was rebranded 'X' in July 2023, Musk stated his fantastical ambition to make the company into 'the biggest financial institution in the world' (in Saul, 2023).

In line with the embedded finance business model, integrating payments within X is envisaged as the ‘up ramp’ for a deeper push into FinTech economies that will be based on ‘the appeal’ of the ‘ease of use’ of financial services for people who ‘increasingly seek simple, holistic, embedded, and direct experiences’ (Townsend, 2021). The ‘basic business logic’ of the embedded finance model, meanwhile, is that ‘the hard work of acquiring customers’ has already been completed and ‘created the opportunity for a zero customer acquisition cost cross-sell’ (Harris, 2019b). As such, the roll-out of integrated monetary and financial services on X is premised on the so-called ‘network effects’ of the platform (see Nowak, 2023); that is, the demand-side scale economies that propel explosive user growth and enclosure on digital platforms.

In contrast with the now prevalent logic of embedded finance, FinTech economies in North America and Europe initially appeared to confirm the late-twentieth century futurism that predicted ‘the new economics of information’ arising from Internet connectivity would lead to ‘the transformation of retail banking’ (Evans and Wurster, 1997). Newly equipped with the Internet, personal computers and search engines, people were envisaged as shopping around more freely and widely for financial products made available on the proliferating websites of disruptive startup providers who competed with banking incumbents on transaction costs (i.e., interest rates, fees, charges). However, regardless of whether startups, banks, BigTechs or other nonfinancial institutions are acting as FinTech intermediaries, the competitive dynamics of FinTech economies have proven to be quite different.

Competition over FinTech intermediation and rent capture turns on assembling or renovating platform infrastructures to enclose the mundane monetary and financial relations of a significant number of people, and/or partnering with others who are similarly pursuing the platform business model (Langley and Leyshon, 2021). Platform intermediation by FinTech intermediaries is also data-driven and data-derived, most obviously when payment firms compete to prospect data rents, for instance (Maurer, 2015; Westermeier, 2020), and when nonfinancial firms leverage their data assets to provide embedded credit products to people without formal credit histories (Breckenridge, 2019; McDonald and Dan, 2021). The platform and data imperatives of FinTech enclosure are such that the logic of intermediation is not to improve how people can access and choose between formal monetary and financial products. This is epitomized by the nesting of ostensibly indispensable and inescapable monetary and financial services within the super apps and digital platform ecologies of the major technology conglomerates. It also frames the intermediary techniques of embedded finance that are under development by a host of nonfinancial companies, including X.

Put another way, it is especially revealing that people in today’s FinTech economies are configured as regular and routine ‘users’ of platformed services rather than as market consumers searching for products. Such ‘configuring’, as Steven Woolgar (1990: 59) puts it, ‘includes defining the identity of putative users, and setting constraints upon their likely future actions’. Receiving surprisingly limited critical attention from social scientists, the seemingly benign description of people as ‘users’ is common throughout digital capitalism and FinTech economies (McNeil, 2020). Users are the essential figures of digital capitalism and FinTech, save for when they are corralled together with their ‘peers’, or rounded up into the collective figure of ‘the crowd’.

The roots of the notion of ‘user’ can be traced to the broad fields of design and engineering, and more recently to microcomputing and ‘the requirements of software and interaction design’ which ‘came to rely on the specification of diverse groups of multiple hypothetical *Users*’ (Bratton, 2016: 255, original emphasis). Also manifest in a plethora of user-related terms and the growth professions of User Experience (UX) and User Interface (UI) design, attracting and – critically – retaining legions of people as user populations is the main competitive objective and metric of digital platform business strategies of

enclosure, including strategies for FinTech intermediation that target active monetary and financial service users.

In present-day FinTech economies and embedded finance, in particular, people are increasingly users of monetary and services that are made available to them within the proprietorial and digital infrastructural platform enclosures which they already inhabit. Marx, of course, placed the combination of materials and labor that is necessary to produce a thing of 'use-value' at the core of capitalist commodity production (Marx, 1990: 125–133). But the user relations of FinTech economies are rentier relations rather than commodity relations. User actions become an important source of data to be aggregated and analyzed for potential revenues (i.e., data rents). Meanwhile, the money rents (i.e., fees, charges, interest) that FinTech intermediaries demand from users are often arbitrarily defined, specific to the platform and/or shrouded in the data-derived differentiations of what is known as personalized and risk-based pricing.

As Sara Ahmed (2019: 6–7) reminds us, 'use' is not a simple category of human action but, by definition, is a 'distributed' and 'instrumental' sociotechnical relation between people and things – such as software and FinTech services. Use is thereby freighted with meaning and affectively charged with 'positive value'. More-often-than-not, use relations are an obligation rather than a choice to be made. In these ways, use is a governmental technique that both differentiates between people and things (who or what is useful? How can it be made useful?) and flattens differences of all kinds (we are all users now) (Ahmed, 2019: 103–130). The depiction of FinTech and embedded finance services as easy, simple, and useful solutions strongly implies a degree of functionality, a kind of practicality and practicability which 'seems to convey an attitude that can be adopted in life' (Ahmed, 2019: 6). The invasive intermediary and rentier operations of FinTech platforms and the ubiquitous services of embedded finance are thus, for the most part, simply taken for granted by the people they enclose. FinTech disappears into the mundane rhythms of technologically enabled socio-economic life, such that the uptake of FinTech services (and the rents they beget) is habitual and regularized. People are used because they make use of embedded finance.

Conclusion

For commentators, consultants, investors and practitioners in North America and Europe, 'embedded finance' has become an increasingly important touchstone in their accounts of FinTech economies. This essay has argued that the rise of embedded finance furthers FinTech disappearance; that is, the erasure of FinTech as a more-or-less distinct market domain of technologically facilitated monetary and financial relations. Consumer-facing FinTech startups were supposed to deploy digital technology to deepen financial market relations, not least by disrupting and disintermediating the business of banking. Yet, specialist FinTech firms are no longer to the fore in North American and European FinTech economies and people are increasingly configured as routine users rather than sovereign consumers of FinTech services. FinTech and especially embedded finance are, in fact, contributing to the dissolution of the final vestiges of marketized monetary and financial relations.

As we have stressed throughout, however, this should not be taken to imply that FinTech economies are fading away or on the decline. Far from it. As Baudrillard (2009: 33) suggests, disappearance is perhaps the ultimate expression of appearance under 'the hegemonic of the digital'. With the assistance of banks and financial institutions, FinTech intermediation and rent capture is increasingly dominated by the monopoly platform enclosures of the leading digital firms. FinTech economies are now marked by the disappearance of the seemingly inescapable intermediary and rentier operations of

FinTech into the institutional and technological background of people's platformed practices of social reproduction and consumption.

It is therefore perhaps tempting to conclude that FinTech economies have been subsumed within digital capitalism and associated tendencies toward 'Techno-Feudalism' (Durand, 2024). After all, people are increasingly dependent on rentier monopoly platforms for their monetary and financial lives. Such a conclusion would serve, however, to once again and mistakenly cast embedded finance and the dynamics of FinTech disappearance as a process of elimination and extinction, albeit in more critical terms. Framed as symptomatic of Techno-Feudalism, it would be easy to overlook how embedded finance and FinTech disappearance propels a penetrating transformation that, so to speak, extends all the way down and into the social relations and subjectivities of everyday money and finance. Registered and furthered as the rise of embedded finance, FinTech disappearance is arguably the fullest expression of the transformative appearance of FinTech in people's monetary and financial lives over the last quarter of a century. Increasingly, wherever people find themselves in a digital world, FinTech has already arrived to find and render them as regularized users of ubiquitous monetary and financial services who are ripe for rent capture.

Notes

1. By March 2025, the sale of the loans at a discount had been gradually completed by the banks, except for around US\$1 billion of the riskiest junior debt that carried an interest rate of 13 percent. To clear this remaining and expensive bank debt, Musk and others purchased further equity in X in a new US\$1 billion funding round that once again valued the company at US\$ 44 billion (Hammond, 2025).
2. For our book-length account of the emergence and development of FinTech economies across the globe, see Langley and Leyshon (forthcoming).
3. We would like to thank Amin Samman for highlighting that 'disappearance' is not only a feature of FinTech that we foregrounded in an earlier conference presentation of the ideas contained in this essay, but also a concept developed for wider application by Baudrillard.
4. See, for example, <https://www.marqeta.com/blog/real-world-examples-of-embedded-finance>.

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